⇒ unsupervised / semi-supervised

↓

How? → K-means / K-mean++

→ DBSCAN

→ PCA / PCA++

⇒ Backprop:

⇒ End-to-end ML pipelines: learning | classification | clustering ...

⇒ Projects...

unsupervised learning

↓

$\vec{x} \xrightarrow{f(\vec{x})} y \rightarrow \in \mathbb{R}^D \longrightarrow \mathbb{R} \rightarrow \mathbb{I}^{[0,1]}$

→ somebody to correct / evaluate

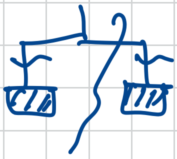
↓

penalize

$f(\vec{x}) = \theta^T x + \epsilon_\theta$

$f(\vec{x}) =$ max min $(\ldots) \rightarrow$ SVR

$f(\vec{x}) =$



many possibilities / outcoms

↓

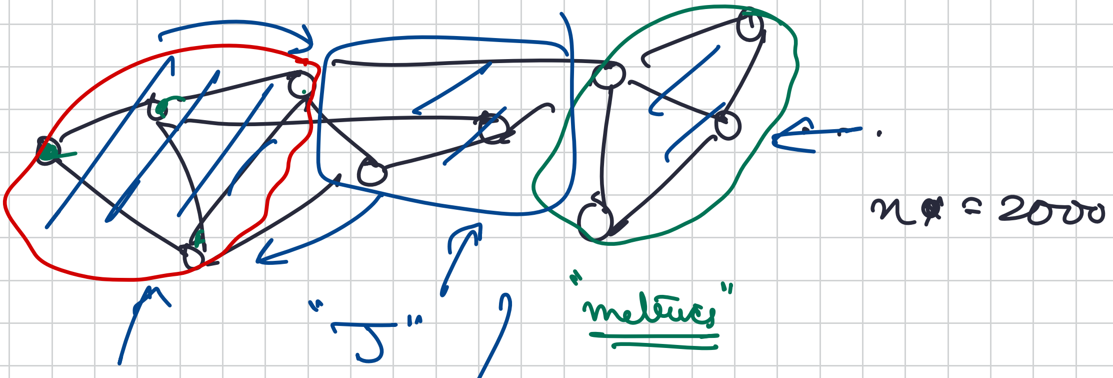
no-one to correct you!

↓

train | testing } → use!

social networks: diff. comm't / Modularity...

↳ graph coloring problem...



→ Locality: ✓ . . K-means

→ Anomaly detection

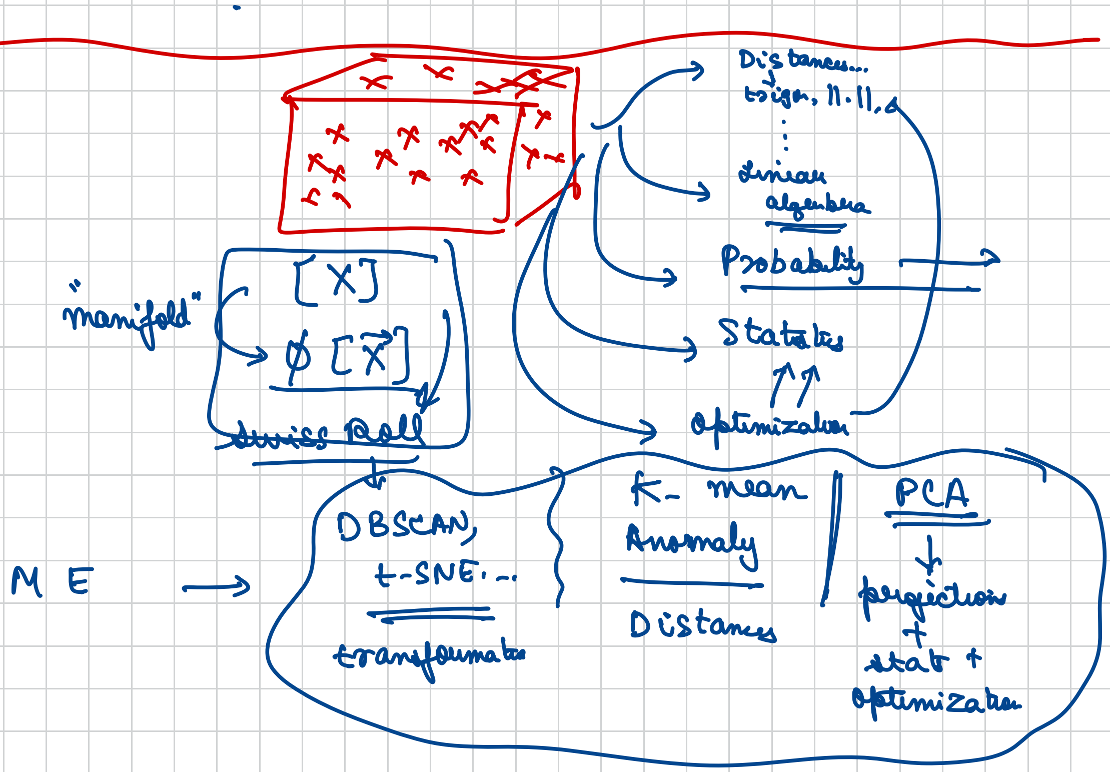
supervised...

AR
ARMA
ARIMA

$$y[k] = \alpha \, y[k-1] + \epsilon_t$$

→ CV → segmentation

⋮



K- mean

K - ?

How many

→ (x_k, y_k)

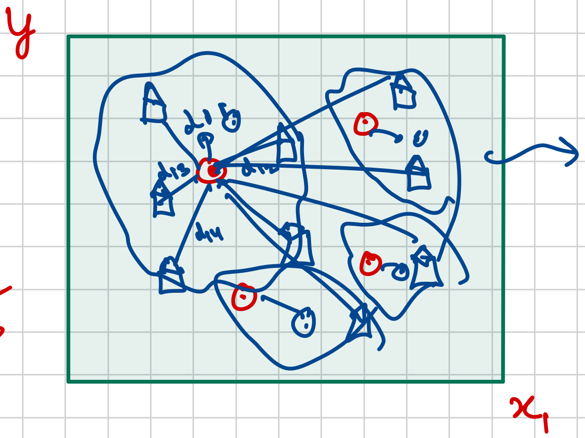
II

$k \in [1, K]$

$K=1$

$K=2$

$K=3$



1000 houses...

K - dark

slows...